



Nursing Practice Guidelines

Intravenous Therapy

Reference #:4

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation -wide	Nursing

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Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to nursing management and administration of intravenous therapy. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☒
Other:	☐		

Principles

It is a legal requirement that all disciplines that prescribe or administer medications comply with the Medicines and Poisons Act 2014, Medicines and Poisons Regulations 2016, Pharmacy Act 2010 and Hospital Guidelines. It is a legislative requirement that a valid prescription must exist before Schedule 4 and Schedule 8 medications are administered¹.

A nurse who is endorsed as a nurse practitioner with the Nurses and Midwives Board of Australia and working in a designated area has the legislative authority to prescribe medications from approved clinical protocols.

All prescriptions must contain:

- Patient identification label
- Approved generic name of the medication
- Dose of medication
- Route of administration
- Rate of administration
- Start date, frequency and prospective date frequency
- Nature and volume of diluents'
- Legible printed surname and signature of medical officer.

All intravenous fluid bags (and bottles) which have had injectable medicines / additives added in the clinical area prior to administration must have an additives label completed and attached to the front of the bag (or bottle) ensuring the fluid name, batch number, expiry and graduations remain visible.² (Image 1).²

Fluid bags and bottles for an infusion where no additional injectable medicines are added prior to administration, examples include intravenous fluids (e.g., 0.9% sodium chloride, 5% Glucose), pre-mixed solutions (e.g. 0.9% sodium chloride with 20mmol potassium) DO NOT require additional labelling)²

The ROUTE OF ADMINISTRATION and DATE AND TIME COMMENCED must be identified on all administration lines (Image 2)²

NOTE: Administration lines connected in an emergency, should be labelled as soon as practicable.²

Image 1: Example of Labelling
Intravenous Fluid Route

Image 2: Example of Label
Intravenous Line - Route

Administration lines to identify medicine: a continuous infusion line that must be retained for delivery of a single medicine must not be used for any other medicine, (e.g. noradrenaline). The medicine contained in the line must be clearly identified alongside the route with both labels placed near the injection port on the patient side (Image 3)²



Image 3: Example of line labelling – route and medicine;

continuous infusion of a single medicine

Patient identification must be checked when matching patient's identity to care, therapy and services.

Observe the five moments for hand hygiene and aseptic non-touch technique. Follow the risk assessment outlined SCGOPHCG Infection Prevention and Management Manual - Aseptic Technique Policy. Appropriate Personal Protective Equipment (PPE) must be worn to prevent contamination and mucosal or conjunctival splash injuries.

1.0 Intravenous Infusion / Additives and Bolus Dose

1.1 Medical Requirements

Intravenous infusions and infusion rates are ordered by medical staff. The patient's ongoing needs for IV fluids must be assessed daily and the management of patient's fluids and electrolytes needs should be included as part of this review^{1,2}. The medical officer must order the medicine dose and rate of infusion and diluent volume on the Fluid Therapy and Intravenous Additives Orders or other endorsed fluid prescription chart (i.e. Variable Fluid Volume Chart).¹

1.2 Safety Information

Independent double checking of the medicine/ fluid/ diluent is required prior to administration. Independent double checking is a procedure that requires two qualified health practitioners, one of whom must be a registered nurse (RN) or medical officer (MO), independently checking the medication³. (If intravenous therapy is being checked by two nurses, one **must be an RN**.^{6,8,9}

Prior to intravenous administration inspect packaging for damage, check expiry dates and sterility indicators.

The therapeutic substance must be prepared immediately prior to use. Any medicine or fluid that cannot be identified (e.g. in an unlabelled syringe or other container) is to be considered unsafe and discarded immediately.²

All intravenous fluids including those with additives must be used within 24 hours of commencement or discarded.

When preparing intravenous infusion or adding additive/ medications observe for signs of incompatibility (i.e., precipitation) – **if present do not continue**, discard all used equipment and start the preparation procedure again.^{6,8}

Administration sets are single use devices and if they are disconnected from the IV device for any reason e.g., intermittent medication dosing, the set is to be discarded and a new administration set connected using aseptic technique.

Additives to infusions

Medications to be administered via an infusion pump

To ensure the accurate administration of medications via infusion pumps, medications must be administered in accurate volumes. This necessitates that all intravenous medications to be administered through an infusion pump are prepared accurately by removing the volume of fluid from the IV fluid bag PRIOR to the addition of the medication.

Patient identification must be checked when matching patient's identity to care, therapy and services.

Consider insertion of CVAD for vesicant / irritant infusate or difficult IV access.⁵

The unnecessary use of IV fluids should be avoided and they should be ceased immediately once they are no longer required.^{4,5}

1.3 Nursing Management

Bedside Checking procedure

- The nurse is responsible for checking known allergies prior to administration of medicines.
- Student RN's may administer intravenous medicines under the direct supervision of a Registered Nurse.
- It is the responsibility of the person checking to⁶:
 - observe the written order,
 - observe the preparation of the medicine,
 - identify the patient at the bedside with the person administering the medicine,
 - check and confirm the rate / dose,
 - observe the initiation of the medicine administration / regulation, and
 - Initial and document on the fluid therapy order chart.
- For Schedule 8 medicines, two qualified health practitioners, one of whom is a RN or MO must check to the bedside and the checker must stay for the duration of bolus administration / commencement of continuous infusion^{3,6}.
- All Schedule 8 medicines must be administered by a RN/ medication competent Enrolled Nurse (EN).^{1,6}
- Before using the N code all reasonable steps should be taken to acquire the medication (through Pharmacy, on call pharmacist after hours via afterhours CNS, from another ward or by contacting the Duty Medical Officer).
- If "W" or "N" codes are used, document the rationale in patient's records.
- If "N" is used, ensure that the medication is ordered from pharmacy by completing the pharmacy medication requests form.
 - OPH- outside OPH Pharmacy opening hours is as per the OPH Hospital Wide After- Hours Procurement of Medicines Guideline
 - SCGH- On the weekend and public holidays: contact the weekend clinical pharmacy service.
- Omitted doses may be administered at a later time; check with the Medical Officer or the ward Pharmacist and document time given.
- Use area specific protocol when appropriate.
- **SCGH only:** Pharmacy Department provides an IV additive service 0830 – 16:30 hours Monday to Friday, 0900 - 1600 hours Saturday, Sunday, and Public Holidays.

a) Intravenous Infusion

- Intravenous fluids will be checked by qualified health practitioners, one of whom must be a RN or MO, independently checking the medication³. Both staff members will initial the fluid therapy and intravenous additive orders
- The use of a burette is to prevent accidental rapid infusion of large fluid/medicine volumes. All intravenous infusions must be connected to a burette, except for:
 - Those going through an infusion pump.
 - Intravenous Antibiotics
 - Non additive intravenous fluids with a volume less than 500ml.
 - Blood products.

Calculating Rate of Flow:

Check manufacturer instructions for the delivery rate (i.e. 20 drops/ mL) of the giving set used for infusion.

Formula:

$$\text{Rate of flow (drops/minute)} = \frac{\text{Volume x Drops per mL}}{\text{Time in Hours x 60}}$$

Or

$$\text{Time [hours]} = \frac{\text{Volume x Drops per mL}}{\text{Rate in Drops / mL x 60}}$$

$$\text{mL / Hour} = \frac{\text{Volume}}{\text{Time (hrs)}} \quad \text{or} \quad \frac{\text{Volume x 60}}{\text{Time (mins)}}$$

b) Additive and Bolus Dose

- Provided a valid prescription exists a RN/ medication competent EN may add any prescribed therapeutic substance to a mini bag including first dose.
- **Exceptions are:**
 - cytotoxic medicines and other medicines, which may be considered to pose an occupational risk. Consult Clinical Pharmacist if unsure.
 - Note: All intraTHECAL medicines/ solutions are obtained from the Pharmacy Aseptic Service Unit, and are only administered by medical staff

1.4 Procedural Information**Requirements**

- Clean disposable tray/kidney dish/trolley
- 2 x 10ml sodium chloride 0.9% Posiflush SP™ syringe
- Water for injection/Sodium chloride injection 0.9%
- 2x 10-20ml syringe (syringe size dependent on medication)
- Drawing up device: 18g drawing up needle or 19g drawing up needle or vial access pin
- Medicine to be administered
- Sterile 70% alcohol impregnated swab
- Medication chart
- Additive label and Line Label
- Appropriate Infusomat® administration line/set
- Personal Protective Equipment (PPE)

PROCEDURE**Preparing Intravenous Infusions**

All preparations must be undertaken using a clean work surface i.e., treatment/ clean utility room or trolley.

1. Perform hand hygiene.
2. Clean the work surface with combined detergent disinfectant wipe. Allow to dry.
3. Gather equipment.
4. Perform hand hygiene.
5. Apply non-sterile gloves if required.
6. Disinfect medicine vial, if applicable, with a 70% sterile alcohol impregnated swab and allow to air dry prior to penetration.
7. Prepare therapeutic substance ordered. Protect key parts using non-touch technique.
8. Attach additive label to prepared substance.

Prior to Administration of Medication

1. Perform hand hygiene.
2. Identify patient by using the 3 points of identification and checking unit number on identification band with medication chart.
3. Check for allergies⁸.
4. Expose IV device i.e., removing dressing/ bandage.
5. Assess integrity of vascular access device prior to administration of medicine. A PIVA Score should be recorded on the Peripheral Vascular Catheter Insertion Record each shift^{5,8,9}. Devices should be stabilised in a manner that enables assessment and monitoring of the cannula site for patency, infiltration or extravasation.
6. Consider insertion of CVAD for vesicant/ irritant infusate or difficult IV access.

Injection

Accessing intravascular device (e.g. peripheral vascular device, CVC, PICC)

1. Perform hand hygiene.
2. Apply non-sterile gloves.
3. Using a non-touch technique, scrub the key part (i.e., port /tip/diaphragm) using a large 70% alcohol wipe to scrub the rubber septum, if applicable for 30 seconds. Use different areas of the wipe. Allow to air dry.
4. Using a non-touch technique flush cannula with 5mls or CVC/PICC with 10mls of sodium chloride 0.9% to determine patency.
5. Inject therapeutic substance over specified time.
6. Flush device with sodium chloride 0.9% as prescribed⁸.
7. Dispose of sharps and equipment appropriately. Remove gloves and perform hand hygiene.
8. Document administration.¹

Injection into IV tubing when infusion in progress:

1. Follow steps 1- 4 as a per accessing intravascular device (above)
2. Pause infusion.
3. Occlude IV infusion tubing.
4. Inject therapeutic substance over specified time.
5. Flush line / tubing with sodium chloride 0.9% as prescribed⁸.
6. Resume infusion.
7. Dispose of equipment appropriately, remove gloves and perform hand hygiene.
8. Document administration.¹

Injection into mini bag/burette:

1. Follow steps 1- 3 as a per for accessing intravascular device (above)
2. Remove the volume of the medication to be added from the fluid bag prior to the addition of the medication.
3. Using a non-touch technique, disinfect the key part (i.e., port /tip/diaphragm) with a large 70% alcohol impregnated wipe and allow to air dry prior to penetration.
4. Inject therapeutic substance into mini bag / burette and agitate solution: Carefully inspect solution and discard if there is any particulate matter, precipitation or change in colour or clarity.
5. Complete additive label and place labels on the front of the mini bag to ensure the name of base fluid, batch number and expiry date remain visible.
6. Document administration¹.

Infusion

Infusion via Mini Bag:

1. Follow steps 1- 4 as a per Accessing intravascular device (above).

2. Manually prime administration line.
3. Prepare and select program on infusion pump.
4. Connect IV line to Vascular Device (or J loop if applicable for device).
5. Prepare and select program on infusion pump
6. Infuse medication and flush device as per prescription.
7. Dispose of equipment appropriately, remove gloves and perform hand hygiene.
8. Document administration¹.

Administration sets are single use devices and if they are disconnected from the IV cannula for any reason e.g., intermittent medication dosing, the set is to be discarded and a new administration set connected using aseptic technique.

For continuous infusions, without disconnection, lines must be changed every 72hrs.

ICU Practice Only

Continuous infusion lines, without disconnection, can remain in use for up to 7 days in ICU when connected to CVC or arterial line with the following exceptions:

- TPN (daily bag change)
- Propofol (changed every 24 hours)
- Blood products (as per transfusion policy)

Continuous infusion lines that have been in use for 72 hours or more must be changed prior to transfer out from the ICU.

2.0 Administration by Enrolled Nurses

2.1 Safety Information

All intravenous therapy is to be **checked by two** qualified health practitioners, one of whom is a RN or MO³.

2.2 Nursing Management

- As per Intravenous Therapy Section 1: Intravenous Infusion /Additives and Bolus Dose.
- An Enrolled Nurse (EN) (*i.e., not registered as a medication competent enrolled nurse*) may care for a patient receiving IV opioid infusion, cytotoxic or epidural/ regional therapies however the preparation and commencement of these therapies is the responsibility of a designated Registered Nurse. In this situation an enrolled nurse is:
 - under the direct supervision of a designated RN,
 - proficient in-patient care relevant to opioid infusion, cytotoxic, epidural or regional medication,

NOTE: Administration of all, cytotoxic medications, or medicines with a low therapeutic index (e.g., infusion and bolus doses) must be performed by the responsible RN.

An Enrolled Nurse may NOT care for unstable patients receiving continuous infusions of high-risk medicines with low therapeutic index that are dependent on monitored medicine levels or dose adjusted/ dependent on frequent patient observation⁹.

3.0 Infusion Pump Guideline

3.1 Medical Requirements

A valid prescription for intravenous therapy including rate of infusion is to be legibly and legally prescribed by medical staff; charted on the intravenous fluid therapy and intravenous additive orders.¹

3.2 Safety Information

All intravenous therapy is to be **checked by two** qualified health practitioners, one of whom is a RN or MO.(3) It is the nurses' responsibility to check the rate of infusion and to ensure the pump is working efficiently at all times.

B Braun Infusomat® Space® infusion pump

The B Braun Infusomat® Space® infusion pump is a volumetric delivery device that is designed to ensure accurate and controlled administration of intravenous fluids, additives and medications¹⁰.

The B Braun Infusomat® Space® infusion pump is supported by Dose Error Reduction Software (DERS), a system of pre-programmed medicine libraries including administration protocols providing safe limits for prescribing and administration¹⁰.

The medicine libraries should always be used except in the following situations:

- Medical emergencies,
- Medication is not in the medicine library – notify the clinical pharmacist and complete the Drug Library Feedback Form available on the intranet.
- Approved medication administration protocol is not included within the medicine library – notify the clinical pharmacist and complete the Drug Library Feedback Form available on the intranet.

Sir Charles Gairdner Hospital (SCGH) will only have those medicine libraries applicable to SCGH and the same for Osborne Park Hospital (OPH).

After turning on the infusion pump and inserting the primed line, the nurse must confirm the required care unit for their ward/ department is displayed

To change the Care Unit:

- Select use of the drug library
- Scroll 'Δ' from 'All drugs' to highlight 'Change Care Unit'
- Press '◀' and select the correct care unit for your area

The following Medicine/ Drug Library Care Units are available for use.

OPH Care Units:

- OPH General Ward Library: for use in Ward 2, 3, 4, 5, 6, Operating Theatres, Radiology
- OPH Chronic Pain (limited to specific pumps only)

SCGH Care Units:

- SCGH General Ward Library- for use in C14, C16, C17, HomeLink, IV Lounge, SSU, G41, G51, G52, G53, G54, G61, G62, G63, G64, G65, G66, G71, G72, G73, G74, G75, Woods HPH, Discharge Ward
- SCGH Critical Care Library- for use in the Emergency Department, Intensive Care Unit, G42, Cardiovascular Interventional Lab, Operating Theatres, Radiology, Nuclear Medicine, NIISwa
- SCGH Haematology/ Oncology- for use in G73, Cancer Centre
- SCGH Chronic Pain Library- for use in the Pain Clinic only

Prior to administering any intravenous medication, two nurses must independently check and confirm:

- Patient identification
- The most up to date version of the correct Care Unit Medicine Library for their area is selected e.g., SCGH General Ward Library.
- Prescription
- Infusion
- Lines

And when transferring a patient between clinical areas two nurses (one of whom is a RN) must check the correct Care Unit Library is in use e.g., transfer from Critical Care to General Ward area.

Infusions and programmes must be checked during temporary or permanent clinical handover to ensure the appropriate Care Unit medicine library is in use. This can be checked whilst the infusion is in progress by scrolling up or down until the Care Unit name is displayed at the base of the running screen.

Flushing with minimum of 30 mL of sodium chloride 0.9% is required post administration of any medicine or additive to clear line of medication.

Standby Mode: The pump can be put into standby mode when a therapy is required to be paused for a period. Always ensure the medication/fluid in the line has been selected in the drug library before using the standby mode.
The maximum standby mode is 24 hours¹⁰.

Infusomat® Space® Line SafeSet:

The administration sets which have the GREEN BAND around the drip chamber incorporate B. Braun Safeset technology. Safeset incorporates two main features – “Airstop”, and “Primestop”

Types of Infusomat® Administration Lines/ sets¹⁰

- Users should refer to consumable packaging to identify if line is Space Line or Space Line Neutrapur.
- Lines are not solely identifiable by the “White” and “Green” banding.
- Check administration set packaging to confirm the total volume required to prime the drip chamber and line.

- Ensure that with all administration sets/lines that an accurate volume to be infused (VTBI) is entered into the pump, taking into account volume (mL) required to prime the lines.

Infusomat® Space® Line SafeSet (15-micron filter) is the “standard” infusion line¹⁰

- The administration sets which have the ‘Green Band’ around the drip chamber incorporate B. Braun SafeSet technology.
- SafeSet incorporates two main features – ‘Airstop’, and ‘Primestop’.
- These sets are manufactured from Neutrapur which is low adsorbing, non-PVC, DEHP free material.
 - Select the *Space Line Neutrap* option on the pump when beginning the infusion.

Infusomat® Space® Line: 200-micron filter e.g. used for blood products¹⁰

- The administration sets with the ‘White Band’ around the drip chamber do not have the “AirStop” filter.
- These lines will need to be carefully primed to prevent air entrainment.
- These sets are manufactured from PVC, DEHP free material.
 - Select the *Space® Line* option on the pump when beginning the infusion.

Use of a Secondary/Piggyback (Neutrapur) Line¹⁰

The Infusomat® Space® Line infusion pump can run a secondary infusion (piggyback) if the two infusions are compatible, and the primary infusion can be paused.

Note: Use a second infusion pump if the primary infusion is not compatible or cannot be paused.

Baxter Intermate® and the Infusor® Portable Elastomeric Infusion System

Baxter Intermate® and the Infusor® Portable Elastomeric Infusion System is a non-electronic portable elastomeric infusion system and are disposable single use infusion

devices. When filled, the device operates with a sustained internal pressure and medications are delivered through a particle filter and a flow restrictor.

Baxter Infusor® is not to be disconnected until fully dispensed and ready for replacement Infusor® unless directed by medical staff.

Administration sets are not to be disconnected for routine care e.g., showering. The devices are designed to provide a continuous flow of medication over the infusion period at the flow rate indicated on the device.

Flow rates are pre-set, and no programming is required.

Osborne Park Hospital Only:

CADD-Solis Ambulatory Infusion Pump- IV opioid infusions must be administered using the CADD-Solis Ambulatory Infusion Pump. Refer to SCGOPHCG Nursing Practice Guidelines Opioid and Ketamine Intravenous Analgesia and Regional Analgesia

3.3 Nursing Management

Consult with pharmacy or medicine information to confirm that medicine / fluid compatibility, concentration, delivery rates and volumes are suitable for piggyback delivery modes.

Infusion pumps should be used when the solution to be administered must be delivered with greater accuracy or at a higher flow rate than can be provided through a manually adjusted gravity infusion set.

High risk medications should always be administered through an infusion pump

Escorting patients:

Patients with intravenous fluids with additives / medication require a nurse escort at all times. If a patient is attending another department without an escort, it is the responsibility of the nurse caring for the patient to ensure there is sufficient intravenous fluids for the duration of transfer.

Relevant documentation (including Intravenous fluid orders) is to accompany the patient and a nurse must be available to receive the patient.

A nurse **is required** to *escort and remain* with the patient when transferred to the Radiology Department between 2230 – 0830 Monday - Sunday (12) and to other non - ward areas after hours and at weekends.

4.0 Specific Medicines - Administration and Monitoring Required

Please refer to individual SCGOPHCG Medication Management Guidelines for information related to the administration and monitoring required for medications including but not limited to:

- Amiodarone
- Adenosine
- Amphotericin B Liposomal (Ambiosome)
- Digoxin
- Dopamine
- Iron (intravenous)
- Insulin: Adult variable rate intravenous insulin
- Naloxone
- Potassium Supplements: Intravenous
- Potassium Supplements: Oral
- Glycerine Trinitrate (GTN)

Related Documents

Legislation

- Health Services Act 2016
- Medicines and Poisons Act 2014, Medicines and Poisons Regulations 2016
- Work Health and Safety Act 2020

Policies:

- WA Department of Health Mandatory Policy:
 - High Risk Medication Policy MP0131/20
 - Clinical Handover Policy MP0095/18
 - Medication Chart Policy MP0078/19
 - State-wide Medicines Formulary Policy MP0077/18
- Hospital Policy:
 - SCGOPHCG Patient Identification
 - SCGOPHCG Medication Management Policy and Procedure
 - SCGOPHCG Clinical Handover - Communicating for Safety
 - SCGOPHCG Clinical Practice Guideline Peripheral Vascular Catheters (PIVC) - Insertion, Maintenance and Removal of

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Authorisation

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